CHAPTER - 1

INTRODUCTION

1.1 Organization Profile

Loginware Softtech Pvt. Ltd is an emerging start up established in the year 2016 and based in Hassan, tier II city of Karnataka State. Loginware is a knowledge-driven company that values cutting edge technology practices and provides comprehensive solutions to help our customers achieve their goals. Loginware is changing the world by changing the way knowledge can be shared. Loginware has the dedicated young minds striving to connect individuals with each other and with technology. Loginware Softtech Pvt. Ltd. is a proactive player covering the full spectrum of software services, from design, development, implementation, Validation, support and corporate training.

Fig 1.1: Company log

Loginware provides solutions that help its customers develop advanced technological products. Thecompany works on various platforms, including embedded systems, application development, and system software support. It is supported by a strong and dedicated team that focuses on delivering high-quality work.

As a complete end-to-end solutions provider, Loginware offers a wide range of services to help customers redesign and improve their businesses. This enables them to adapt and compete effectively in a rapidly changing market. The main objective of the company is to give its clients a competitive advantage in the marketplace.

1.1History

Over the past two decades, the world has experienced rapid growth in information and communication technology. Tier II and Tier III cities have played a significant role in the socio-economic development of the country. In India, these cities are no longer dependent on traditional methods to run businesses and are increasingly adopting modern technologies.

Loginware Softtech Pvt. Ltd. is an emerging startup based in Hassan, a Tier II city in Karnataka. The vision of the company has been developed over the past two years through a collaborative process. This process involved engineers, academic faculty, and administrative members. It also included the review of various existing platforms and support from different technology organizations.

1.3 Vision

To be a leading technology company, transforming creative ideas into reality.

1.4 Mission

Bringing out the best in everyone we touch; to motivate, inspire, and empower each other to do things they never thought were possible.

1.5 Objectives

The primary purpose of the company is to utilize a carefully designed and integrated set of training programs, entrepreneurship development components, cooperative concepts, and technology services to select and train individuals of Tier-II and Tier-III cities, helping them become a new generation of skilled workers, entrepreneurs, and community builders.

The core objectives carried out at Loginware include:

- Knowledge Hub Creation: Creating a one-stop technology platform and knowledge hub to support all kinds of technological needs of Tier-II and Tier-III cities.
- Facility Integration: Making available at one location all essential technology facilities, training, and support services needed by individuals in these locations.
- Enterprise Development: Systematically creating a substantial number of new modern small business enterprises and worker cooperatives.
- Socio-Economic Growth: Generating economic growth and employment opportunities for youth and underprivileged individuals in Tier-II and Tier-III cities.

1.6 Services

a) Embedded Engineering

Embedded systems and software play a major role in modern life. With the advancement of hardware technology, devices have become more powerful and cost-effective. As a result, embedded software is widely used in various fields such as consumer electronics, transportation, healthcare, and manufacturing.

Loginware provides embedded engineering solutions that support the design, development, and implementation of reliable and efficient embedded systems.

b) Web and Mobile Services

The rapid growth of mobile devices and internet technologies has created an “always-on” or “ubiquitous” society, where digital services are accessible anytime and anywhere. Web and mobile applications have significantly changed the way people shop, work, and live.

Loginware’s Web and Mobile Services group offers solutions that meet both individual and business needs. The major services include:

- Website development and maintenance
- Web-based application development
- Custom software development
- Mobile application development

c) Technology Consulting

In today’s world, business growth and technological innovation are closely connected. Technologies such as Social-Media, Mobility, Analytics, and Cloud Computing have transformed the business environment.

Loginware provides technology consulting services to help businesses adopt new technologies effectively. These services focus on implementing new applications, platforms, and system architectures. The company also helps improve business processes, workflows, and management systems to ensure a smooth transition during technological changes.

Fig 1.2: Services

1.7 Current Running Projects

Fig 1.3: Industry 4.0 Products

a) EAGLE

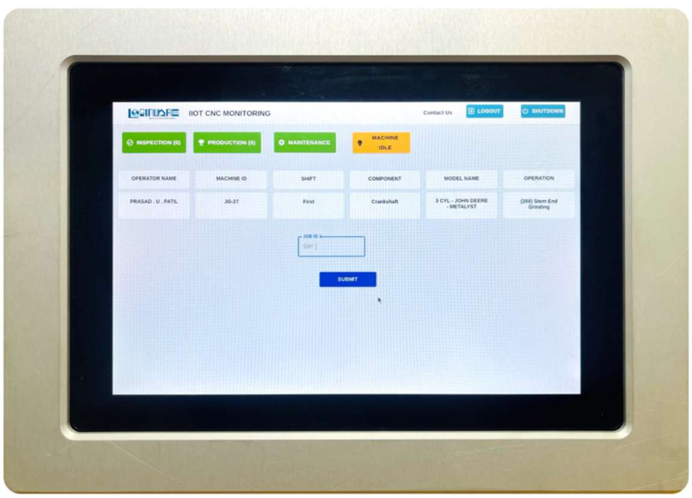

Fig 1.4: EAGLE: Attractive Look with Captivating Features

EAGLE is a powerful embedded hardware solution designed to collect data from various machines, equipment, and sensors. It enables efficient process control and helps reduce unmonitored losses in industrial environments. With its intuitive user interface, EAGLE provides a smooth user experience, allowing operators to focus on their core tasks.

Key Characteristics

- Sleek and modern design
- Compact size for easy integration into factory environments
- Durable and robust construction for long-term reliability
- User-friendly interface with an intuitive touchscreen display
- Advanced connectivity options for seamless integration with manufacturing systems
- Supports multiple communication protocols for real-time data acquisition and transmission
- Easy installation and operation in industrial settings

Key Features:

- Advanced embedded hardware
- 10 Digital Inputs
- 5 Digital Outputs
- 2 Analog Voltage Sensor Inputs
- 2 Analog Current Sensor Inputs
- GPS Integration
- GSM Integration
- Internet connectivity via SIM
- Operating System running on SSD
- Attractive Graphical User Interface (GUI)

b) SPARROW

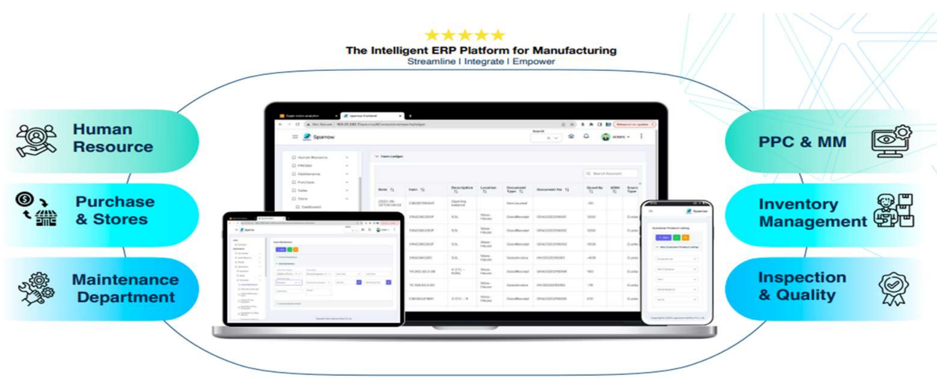

Fig 1.5: Loginware SPARROW

Streamlined Production and Agile Resource Optimization Workflow is a powerful Manufacturing Operations Management (MOM) software developed by Loginware. It is designed to streamline and manage manufacturing and service operations from start to finish. The software supports the efficient planning, monitoring, and control of production processes. It helps improve resource utilization and enhances overall productivity in industrial environments.

c) Analytics

A comprehensive analytics platform that serves as the backbone of the company's integrated manufacturing solutions. It collects and combines data from systems such as EAGLE and SPARROW. By integrating this data, Loginware Analytics enables manufacturers to perform effective analysis and make informed, data-driven decisions. This leads to improved operational efficiency, reduced losses, and enhanced overall performance.

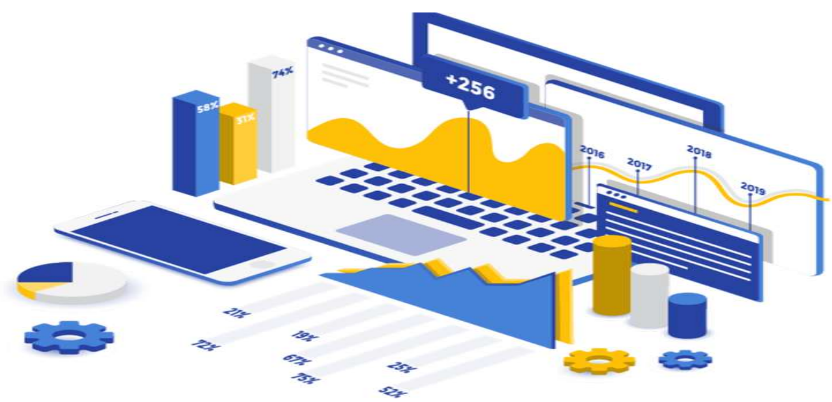

Fig 1.6: Loginware Analytics

Organizational Flow Chart

The structural hierarchy of the organization is oriented toward technical research, development, and training, ensuring smooth execution of projects and efficient operation:

A [Managing Director] --> B [Director Technical]

B --> C [Chief Engineers / R&D Heads]

C --> D [Superintending Engineers / Project Leads]

D --> E [Engineers / Developers & Trainers]

E --> F [Interns / Trainees]

CHAPTER - 2

Narration

2.1 Overall Internship Details:

2.1.1 Phase 1: Foundation and Microcontroller Programming (Days 1 – 48)

The internship program commenced on February 2, 2026, with an orientation session at Loginware Softtech Pvt. Ltd. The initial phase focused on understanding the corporate environment, ongoing R&D projects, and Industry 4.0 practices followed by the organization. During this period, we built a strong foundation in Embedded C programming.

Under the guidance of technical trainers, we worked with the ESP8266 NodeMCU. We configured General Purpose Input/Output (GPIO) pins to process signals from various peripherals. We interfaced digital sensors such as the DHT11 (temperature and humidity), fire sensors, and gas sensors, along with analog sensors like the MQ-3 (alcohol sensor).

Additionally, we implemented Pulse Width Modulation (PWM) techniques to control DC motors using the L293D motor driver. Alarm systems were also developed using relays and buzzers, enabling us to understand real-time embedded control systems.

2.1.2 Phase 2: Python Programming and Edge Systems (Days 49 – 70)

After completing the Embedded C phase, we transitioned to Python programming. This phase focused on learning Python syntax and libraries required for hardware and network-based applications on Linux-based single-board computers.

Training included working with the Raspberry Pi Compute Module 4 (CM4) IO Board. We configured communication protocols such as I^2C, SPI, and UART for device interfacing. We processed data from ultrasonic sensors, displayed outputs on character LCDs, and controlled high-power relays.

Using this knowledge, we developed and tested several real-time projects, including:

- Smart Industrial Safety and Monitoring System
- Smart Supermarket Project
- Multi-Functional Vehicle Security and Driver Safety System

We also integrated IoT capabilities by connecting systems to cloud platforms such as Blynk for real-time monitoring via mobile devices.

2.1.3 Phase 3: STM32 Microcontroller Platform and Skill Strengthening (Days 71 – 90)

In the final phase, training shifted to the ARM Cortex-M architecture using STM32 microcontrollers and STM32CubeIDE. We learned about low-level architecture, clock configuration, and register-level operations.

Our initial task involved setting up the STM32 board and executing a basic LED blinking program to understand initialization and code generation. Subsequently, we rebuilt earlier sensor interfacing projects using Embedded C on the STM32 platform.

This phase significantly strengthened our programming, debugging, and peripheral interfacing skills.

The internship concluded on May 9, 2026, with a valedictory function, where internship completion certificates were awarded. Mentors also shared valuable insights on industry standards and career opportunities in embedded systems.

2.2 Projects Developed During the Internship

2.2.1 Smart Industrial Safety and Employee Monitoring System

➢ Introduction

The Smart Industrial Safety and Employee Monitoring System is an IoT-based real-time monitoring solution developed using Raspberry Pi 4, environmental sensors, RFID technology, and the Ubidots cloud platform. The system is designed to enhance industrial safety by continuously monitoring environmental conditions such as gas leakage and fire hazards, while also implementing employee identification and access control.

Industrial environments are exposed to risks like gas leaks, fire accidents, overheating, and unauthorized access. Manual monitoring can cause delays in detecting such hazards. To overcome these challenges, this system provides automated monitoring, immediate local alerts, and remote cloud-based supervision. The project integrates embedded hardware and IoT technology to create a reliable and efficient safety solution.

➢ Components Used

Hardware Components:

- Raspberry Pi 4 – Central processing and control unit
- MQ Series Gas Sensor – Detects harmful gases and smoke
- Flame Sensor – Detects fire or flame presence
- RFID Module – Used for employee identification
- RFID Tags/Cards – Assigned to employees for access verification
- Relay Module – Controls electrical devices during emergencies
- LED Indicators – Visual warning signals
- Buzzer – Audio alert system
- Breadboard – Circuit prototyping
- Jumper Wires – Electrical connections

Software & Platform Used:

- Python Programming Language – For system logic and sensor data processing
- Ubidots IoT Platform – For real-time cloud monitoring and data visualization

➢ Block Diagram

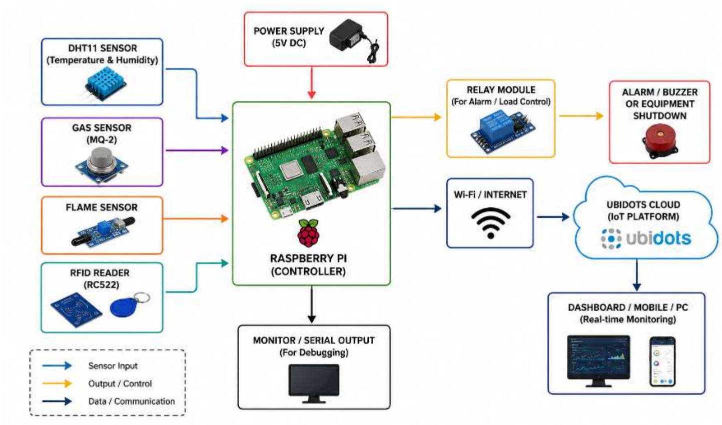

Figure 2.1: Block Diagram of Smart Industrial Safety and Employee Monitoring System

➢ Hardware Implementation

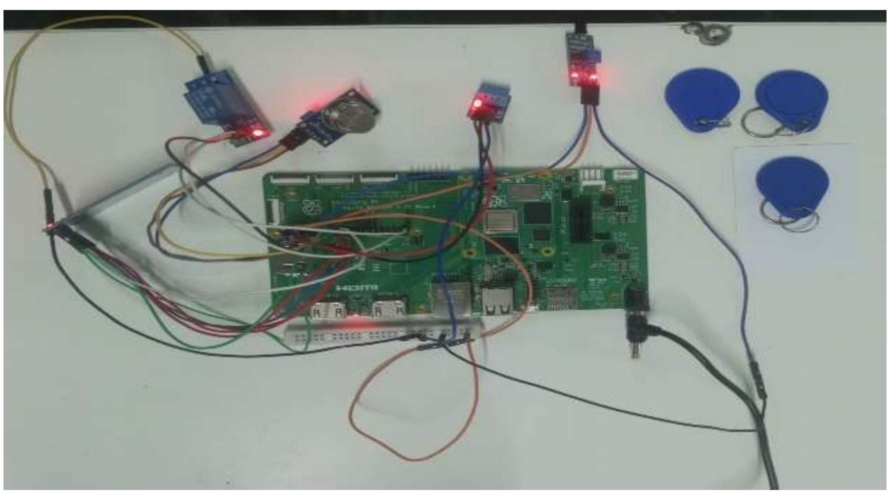

Figure 2.2: Hardware Setup of Industrial Safety System

➢ Working of the System

The system operates continuously to monitor industrial safety conditions in real time.

1. The gas sensor continuously checks air quality and detects harmful gases or smoke.

2. The flame sensor detects fire by sensing infrared radiation.
3. The RFID module scans employee cards to verify authorized access.
4. All sensor data is sent to the Raspberry Pi 4, which processes the information using Python programming.
5. If hazardous conditions are detected:
    - The LEDs turn ON.
    - The buzzer activates to generate an alarm.
    - The relay module can disconnect connected equipment for safety.
6. Sensor data is transmitted to the Ubidots cloud platform through internet connectivity.
7. The industrial conditions can be monitored remotely using the cloud dashboard in real time.

This system ensures automated safety monitoring, instant alert generation, and remote accessibility.

➢ Results Obtained

The system was successfully implemented and tested. The following results were observed:

- The gas sensor successfully detected smoke and harmful gases.
- The flame sensor accurately detected fire conditions.
- The RFID system worked properly for employee identification and access verification.
- LEDs and buzzer provided immediate warning alerts during hazardous situations.
- Sensor data was successfully transmitted to the Ubidots IoT dashboard.
- Real-time monitoring was achieved through cloud connectivity.
- The Raspberry Pi effectively controlled the entire system.

➢ Table: Sensor Readings Under Normal and Abnormal Conditions

Parameter	**Normal Condition**	**Abnormal Condition (Gas/Fire Detected)**
Temperature	25–30 °C	Above 40 °C
Humidity	40–60 %	Below 30 % (or sudden change)
Gas Indicator	OFF (0)	ON (1)
Flame Indicator	OFF (0)	ON (1)
LED Indicator	OFF	ON
Relay Status	ON (System Running)	OFF (Safety Mode Activated)

Table 1.1 Sensor Readings

➢ Ubidot’s Dashboard Realtime readings

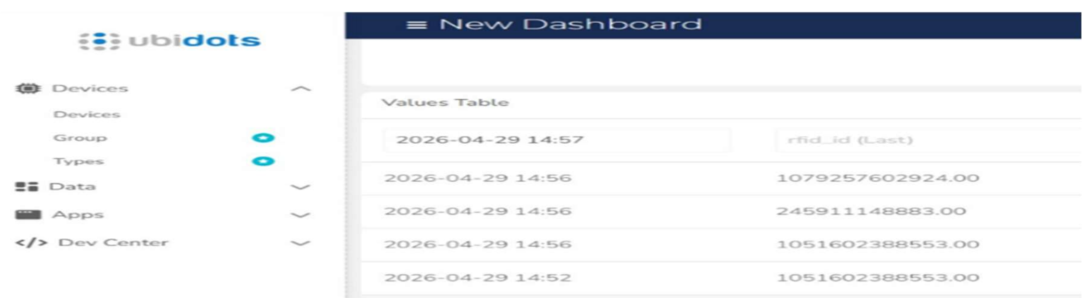

Figure 2.3: RFID Card Reading Displaying Employee ID Number for Authentication

Figure 2.4: Ubidots Dashboard Showing Sensor Monitoring

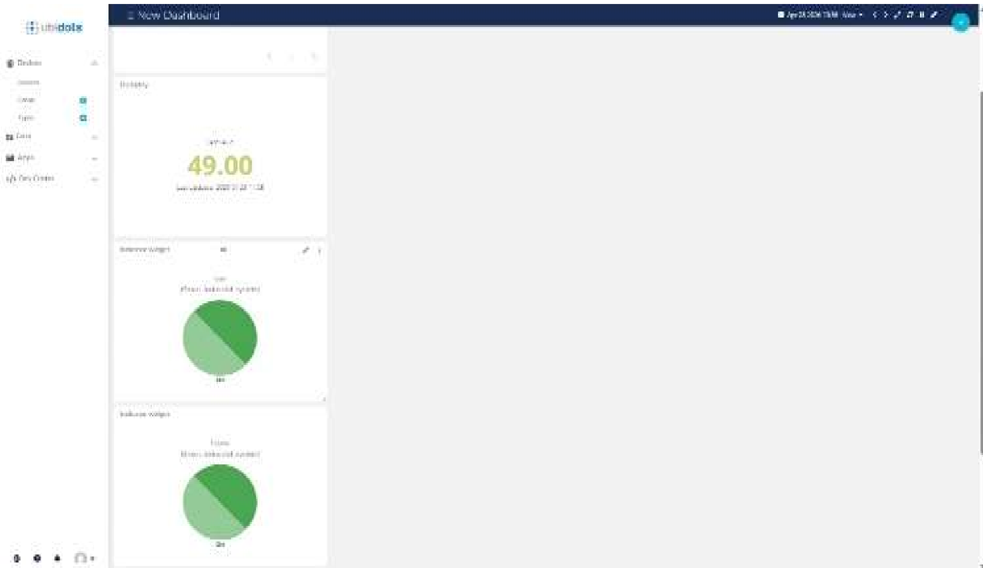

Figure 2.4: Ubidots Dashboard Showing Sensor Monitoring

```
Sensor Data
-------------------------
Temperature : 27.0 °C
Humidity    : 46.0 %
Gas         : ON
Flame       : ON
Response    : 200
-------------------------
Scan RFID Card...
Card ID : 1079257602924
Name    : shikha
Sending to Ubidots: {'rfid_id': {'value': 1079257602924, 'context': {'name': 'shikha'}}, 'temperature': 0, 'humidity': 0, 'gas': 1, 'flame': 1}

Sensor Data
-------------------------
Temperature : 0 °C
Humidity    : 0 %
Gas         : ON
Flame       : ON
Response    : 200
-------------------------
Scan RFID Card...
Card ID : 1041392141900
Name    : russian embassy
Sending to Ubidots: {'rfid_id': {'value': 1041392141900, 'context': {'name': 'russian embassy'}}, 'temperature': 27.0, 'humidity': 46.0, 'gas': 1, 'flame': 1}

Sensor Data
-------------------------
Temperature : 27.0 °C
Humidity    : 46.0 %
Gas         : ON
Flame       : ON
Response    : 200
-------------------------
```

Figure 2.5: Real-Time Sensor Output Display

2.2.2 Smart Supermarket Billing System using RFID

➢ Introduction

The Smart Supermarket Billing System is an RFID-based embedded system developed using Raspberry Pi to automate the billing process in supermarkets. The system scans product RFID tags, displays the selected item details including name, quantity, and cost, and automatically calculates the total amount.

The system also uses a balance card for payment. When the balance card is tapped, the system checks the available balance. If sufficient balance is available, the amount is deducted automatically. If the balance is insufficient, the system displays an "Insufficient Balance" message and allows the user to add a top-up amount. This project demonstrates real-time billing automation using Embedded Systems and RFID technology.

➢ Components Used

Hardware Components:

- Raspberry Pi (Main Controller)
- RFID Reader Module
- RFID Product Tags
- RFID Balance Card
- LCD Display
- Buzzer (Optional for alerts)
- Relay Module (If used)
- Power Supply
- Jumper Wires and Breadboard

Software Used:

- Python Programming
- GPIO Library for Hardware Control

➢ Block Diagram

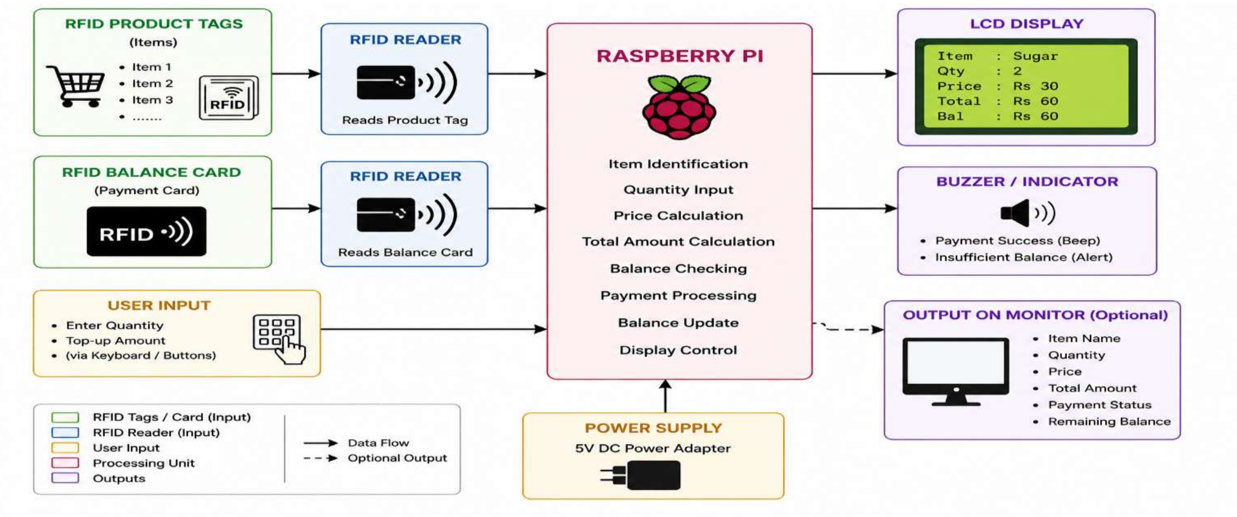

Figure 2.6: Block Diagram of RFID-Based Smart Supermarket System

➢ Hardware Implementation

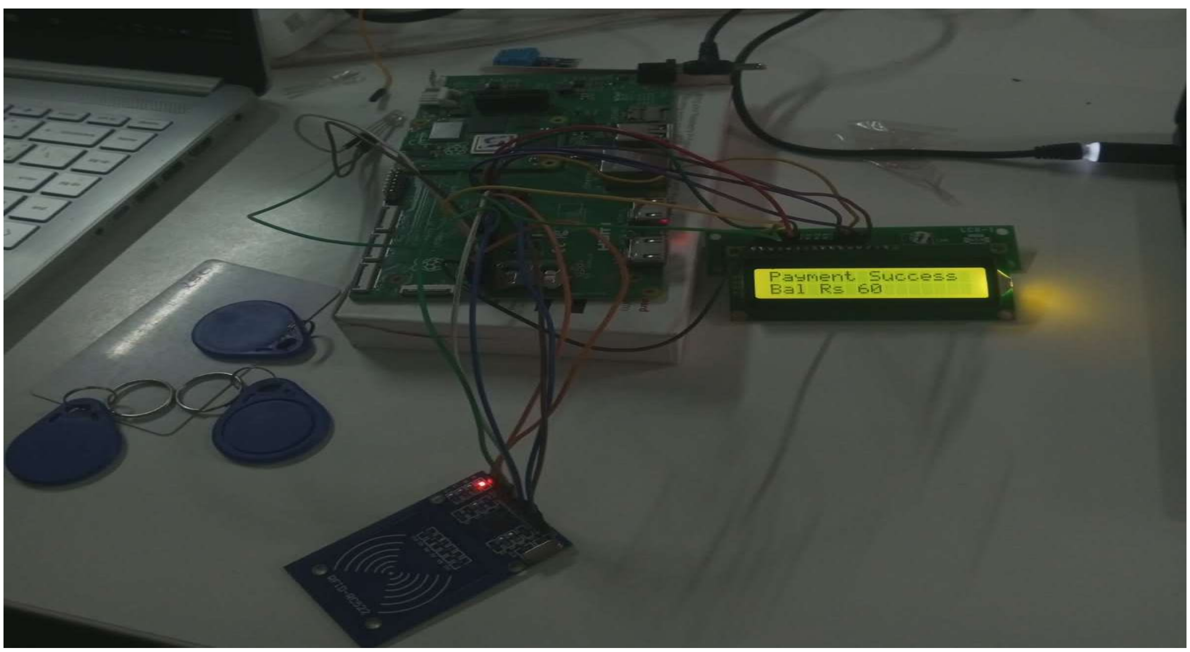

Figure 2.7: Hardware Implementation of the Smart Supermarket Billing System

➢ Working

1. The RFID reader scans the product tag when the item is placed near it.
2. The system identifies the item name from the RFID data.
3. The user enters the quantity.
4. The system calculates the total cost based on item price and quantity.
5. The total amount is displayed on the output section (LCD/monitor).
6. The user taps the balance card for payment.
7. The system checks the available balance in the card.
8. If balance is sufficient:
    - The amount is automatically deducted.
    - The remaining balance is updated.
    - “Payment Success” is displayed.
9. If balance is insufficient:
    - The system displays “Insufficient Balance”.
    - The user can add a top-up amount.
    - After top-up, the amount is deducted and updated balance is shown.

This enables a fully automated and cashless supermarket billing system.

➢ Results Obtained

The Smart Supermarket Billing System was successfully tested and operated correctly.

- Items were scanned accurately using RFID tags.
- The system displayed the selected item name, quantity, and price.
- The total cost was calculated automatically.
- The balance card was detected properly.

- If sufficient balance was available, the payment was deducted successfully.
- If balance was insufficient, the system displayed an error message.
- After adding top-up, the balance was updated correctly.
- The final remaining balance was shown on the system.

The project successfully demonstrated automated billing, balance management, and real-time transaction processing using Raspberry Pi and RFID technology.

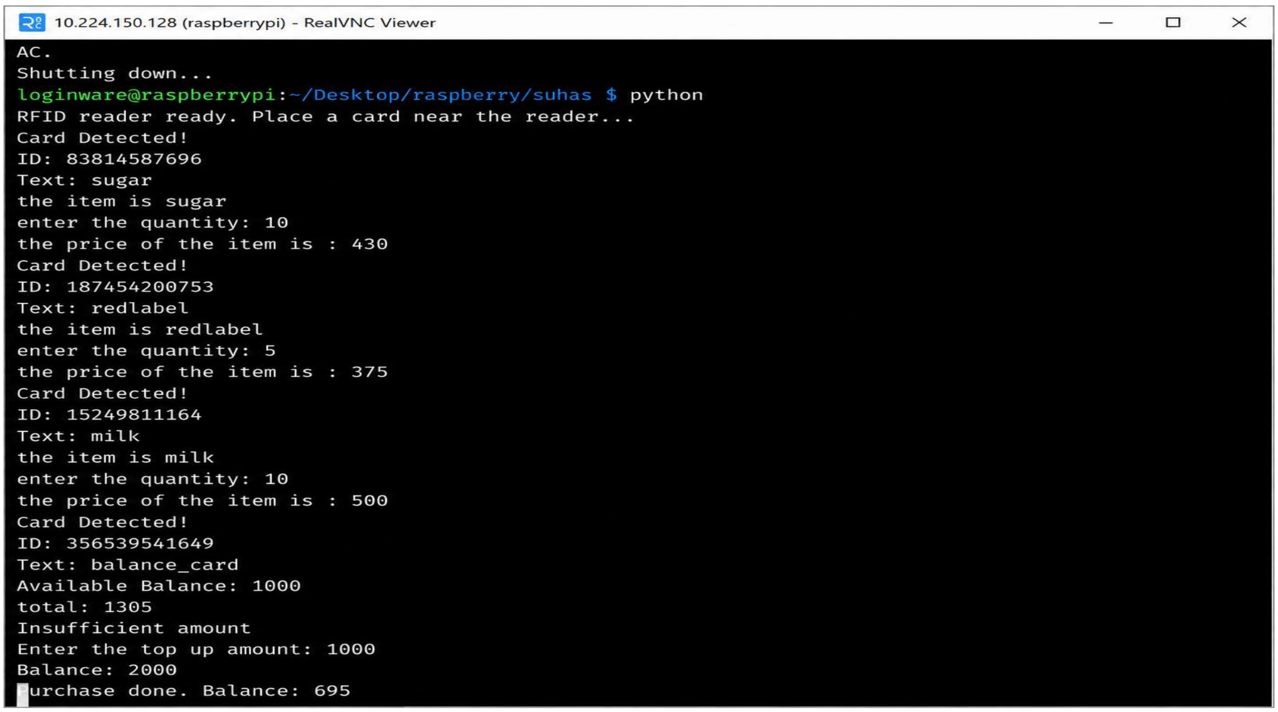

Figure 2.8: Real-Time Output of the Smart Supermarket Billing System After Hardware Implementation

CHAPTER - 3

Schedule of the Internship

Total Duration: 13 Weeks (90 Days)

Start Date: February 2, 2026

End Date: May 9, 2026

3.1 Weekly Schedule of Activities

Week 1 (Feb 2 – Feb 7): Orientation and Foundation

- Introduction to corporate structure and industrial operations
- Overview of products such as EAGLE edge device and SPARROW software
- Revision of C programming and Arduino IDE setup

Week 2 (Feb 8 – Feb 14): Microcontroller Architecture & GPIO

- Study of ESP8266 Node-MCU architecture
- GPIO configuration for input/output operations
- LED control, button interfacing, and buzzer programming

Week 3 (Feb 15 – Feb 21): Serial Communication

- Study of UART, I^2C, and SPI protocols
- Serial communication and AT command handling

Week 4 (Feb 22 – Feb 28): Sensor Interfacing – I

- Interfacing DHT11 sensor
- Reading and displaying temperature/humidity data
- MQ-3 sensor testing and calibration

Week 5 (Feb 29 – Mar 6): Sensor Interfacing – II

- Fire and gas sensor interfacing
- Alarm system implementation using buzzers

Week 6 (Mar 7 – Mar 13): Motor Drivers & Actuators

- L293D motor driver operation
- DC motor control using PWM

Week 7 (Mar 14 – Mar 21): Distance Sensors

- Ultrasonic sensor interfacing (HC-SR04)
- IR-based obstacle detection

Week 8 (Mar 22 – Mar 28): Raspberry Pi & Python

- CM4 IO Board overview
- Linux setup and Python basics

Week 9 (Mar 29 – Apr 4): Advanced Peripherals

- LCD and display interfacing
- Relay control using Python

Week 10 (Apr 5 – Apr 11): IoT & Cloud

- Blynk cloud integration
- Real-time data monitoring
- Completion of safety system project

Week 11 (Apr 12 – Apr 18): STM32 Introduction

- ARM Cortex-M basics
- STM32CubeIDE setup
- LED blink program

Week 12 (Apr 19 – Apr 25): STM32 Programming

- Sensor interfacing using STM32
- Debugging and HAL programming

Week 13 (Apr 26 – May 9): Finalization

- Code optimization
- Report preparation
- Valedictory function and certification

CHAPTER - 4

Overall Experience of the Internship Programme

The 90-day internship at Loginware Softtech Pvt. Ltd., conducted from February 2, 2026, to May 9, 2026, provided an excellent opportunity to bridge the gap between theoretical knowledge and real-world applications in the field of Embedded Systems and Industrial Internet of Things (IIoT). The program offered valuable exposure to Industry 4.0 concepts, enabling us to apply academic learning to practical and scalable industrial solutions.

4.1 Technical Skill Acquisition

The internship was well-structured to gradually develop our skills across hardware, software, and cloud-based platforms.

- **Embedded C and Microcontroller Foundations:** The program began with strengthening our fundamentals in Embedded C programming and microcontroller architecture. Through hands-on work with the ESP8266 NodeMCU and sensors such as DHT11, MQ-3, flame sensors, and ultrasonic modules, we developed strong skills in hardware interfacing, interrupt handling, and data acquisition.
- **Python Programming and Edge Computing:** In the next phase, we learned Python programming using the Raspberry Pi Compute Module 4 (CM4). This helped us understand embedded Linux systems and edge computing concepts. We configured communication protocols such as I²C, SPI, and UART, and gained practical experience in IoT cloud platforms, real-time data visualization, and remote monitoring systems.
- **ARM Cortex-M Architecture:** In the final phase, we worked with STM32 microcontrollers using the STM32CubeIDE toolchain. This provided deeper insights into 32-bit ARM Cortex-M architecture, peripheral driver development, and embedded C code optimization.

4.2 Professional Growth and Industry Readiness

In addition to technical skills, the internship also provided exposure to industry practices such as system design, prototyping, testing, and documentation.

During the internship, we successfully developed several real-time projects, including:

- **Smart Industrial Safety System:** Environmental monitoring for hazardous conditions
- **Smart Supermarket Inventory System:** Automated product tracking and workflow management

- **Multi-Functional Vehicle Security System:** Integrated automobile safety system using sensors and communication modules

The valedictory session held on May 9, 2026, allowed us to reflect on our learning experience. Interaction with R&D engineers and feedback from mentors helped improve our confidence and professional skills.

Overall, this internship has prepared us to handle real-world challenges in the embedded systems domain and has strengthened our foundation for a successful engineering career.

Conclusion

The internship at Loginware Softtec Pvt. Ltd. provided valuable practical exposure in embedded systems and electronics engineering. During the internship, I gained hands-on experience with microcontrollers and development boards such as Arduino Uno, Raspberry Pi, ESP32, ESP8266 (NodeMCU), and STM32F407VGT6. I worked on interfacing various sensors, communication protocols, LCD displays, and embedded applications, which helped strengthen my technical knowledge and practical implementation skills.

The training enhanced my understanding of embedded programming, hardware interfacing, circuit troubleshooting, and real-time applications. I also improved my problem-solving abilities, teamwork, technical documentation, and professional communication skills while working on different tasks and projects.

Overall, the internship was a highly beneficial learning experience that bridged the gap between academic knowledge and industry practices. It increased my confidence in working with embedded systems and motivated me to further develop my skills in electronics, automation, and embedded technologies for future career opportunities.